WEEK 10: STATISTICS III

The final two methods of data presentation:

- Pie Charts → used to show proportions (parts of a whole)
- Frequency Polygons → used to show the shape of a distribution

Statistical Graph Summary

- Bar Chart → discrete data
- Histogram → continuous grouped data
- Pie Chart → proportions
- Frequency Polygon → distribution shape

A. PIE CHARTS

Definition

A pie chart is a circular diagram divided into sectors. Each sector represents a proportion of the whole.

Important Idea

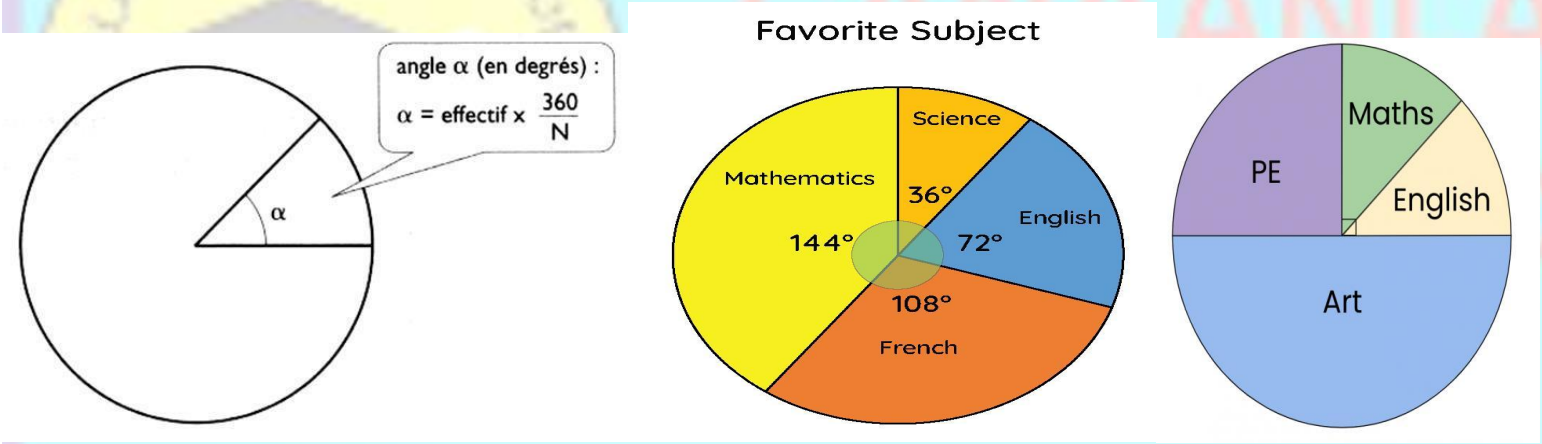
- Whole circle = 360°

- Total frequency = 100%

Formula

Angle of sector = (frequency ÷ total frequency) × 360°

Visual Meaning



Worked Example 10.

An SS1 class has:

- Arts = 30 students
- Science = 40 students
- Commercial = 10 students

Total = 80 students

Step 1: Find angles

Arts:
$(30 \div 80) \times 360° = 135°$

Science:
$(40 \div 80) \times 360° = 180°$

Commercial:
$(10 \div 80) \times 360° = 45°$

Step 2: Check total

$135° + 180° + 45° = 360°$ ✔

Final Table

- Arts = 135°
- Science = 180°
- Commercial = 45°

B. FREQUENCY POLYGONS

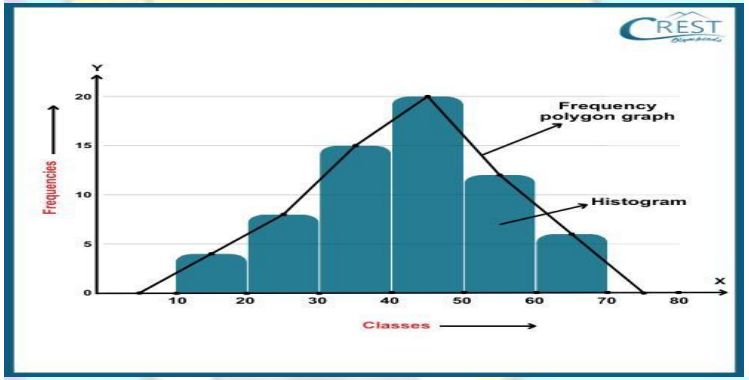
Definition

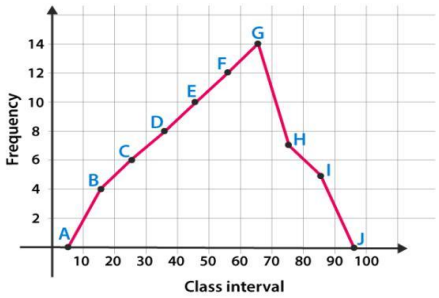
A frequency polygon is a line graph formed by joining the midpoints of class intervals.

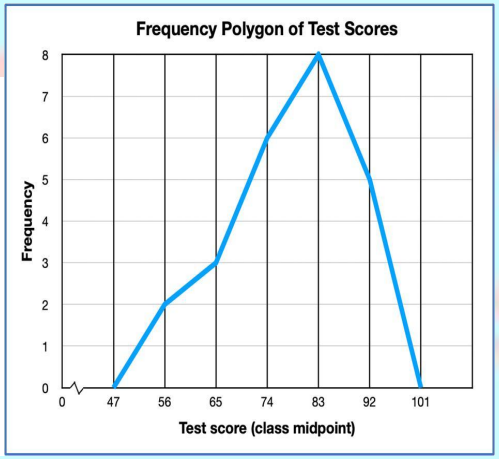
Important Features

- X-axis = class midpoint
- Y-axis = frequency
- Graph is drawn by joining points with straight lines
- Must start and end at zero frequency

Visual Concept







Worked Example 10.2

Height data:

150–154 = 6
155–159 = 12
160–164 = 18
165–169 = 10
170–174 = 4

Step 1: Find midpoints

150–154 → 152
155–159 → 157
160–164 → 162
165–169 → 167
170–174 → 172

Step 2: Add grounding points

Before first class:
147 → 0

After last class:
177 → 0

Step 3: Plotting table

147 → 0
152 → 6
157 → 12
162 → 18
167 → 10
172 → 4
177 → 0

3. REVISION QUESTIONS

1. What is the purpose of a pie chart?
2. What is the formula for angle of a sector?
3. What is plotted on the x-axis of a frequency polygon?

4. What is plotted on the x-axis of a histogram?
5. Why must a frequency polygon start and end at zero?

4. Summary

1. (20 ÷ 80) × 360° = 90° ✔
2. Frequency polygon shows shape more clearly
3. Join midpoints of histogram bars
4. Food = 120° → (120 ÷ 360) × 100 = 33.3%
5. Midpoint of 30–39 = (30 + 39) ÷ 2 = 34.5

5. ASSIGNMENT

1. Family Budget

Total = 30,000 + 20,000 + 15,000 + 10,000 + 15,000 = 90,000

Angles:

Food:
(30,000 ÷ 90,000) × 360° = 120°

Rent:
(20,000 ÷ 90,000) × 360° = 80°

School Fees:
(15,000 ÷ 90,000) × 360° = 60°

Transport:
(10,000 ÷ 90,000) × 360° = 40°

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Savings:
(15,000 ÷ 90,000) × 360° = 60°

2. Frequency Polygon

- Use Week 9 ages data
- Find class midpoints
- Add 0 frequency points at both ends
- Plot and join points

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